Brain and heart

THE BODY'S MOST VITAL ORGANS ARE THE BRAIN AND THE HEART.

The brain and the heart are the most important parts of the body. While the heart is the engine that keeps the body supplied with nutrients, the brain is the control center.

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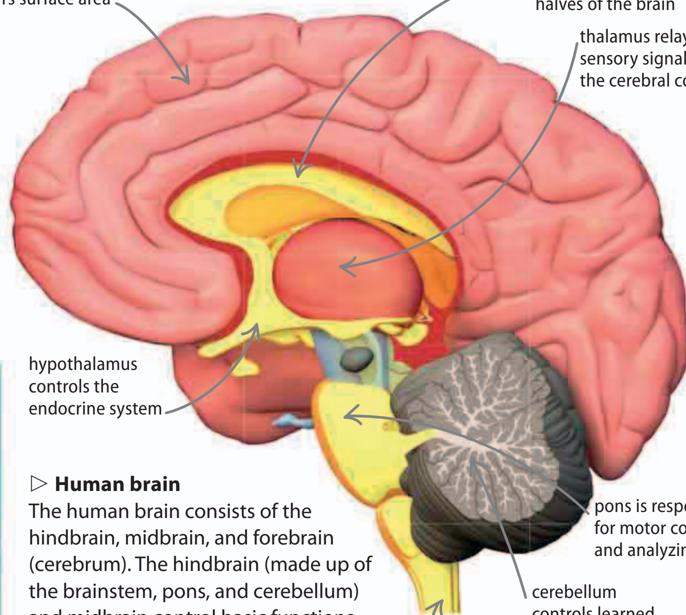
Brain

The brain forms the main part of the central nervous system (CNS), which receives signals from every part of the body, and sends out responses if necessary. The brain is split into two halves, or hemispheres, made of masses of nerve cells that have thousands of high-speed connections with their neighbors. The outer layer of the brain is called the cerebral cortex, or gray matter, and the inner layer is called white matter.



▷ **Human brain**
The human brain consists of the hindbrain, midbrain, and forebrain (cerebrum). The hindbrain (made up of the brainstem, pons, and cerebellum) and midbrain control basic functions, such as breathing and balance, while the forebrain—especially large in humans—is used for thinking and making decisions.

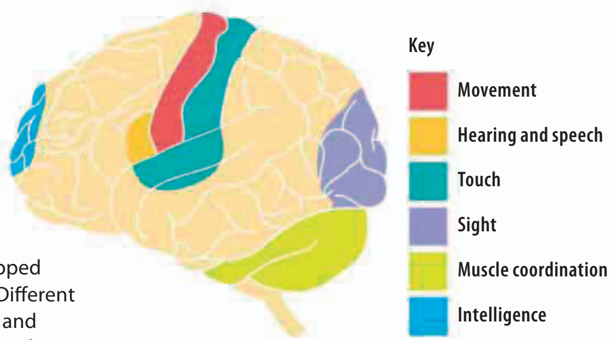
REAL WORLD

Magnetic resonance imaging (MRI)

An MRI scanner causes soft body tissues, such as the brain, to release radio waves for a split second. These are used to build a detailed picture of internal tissues, and help doctors diagnose and treat illnesses.

Brain functions

Neuroscience, the study of the brain, has found that different areas of the cerebrum are devoted to specific functions. If one of the areas—often known as a cortex—is damaged, that function, such as speech or sight, ceases while the others continue unaffected. Neuroscientists have learned a lot about the human brain in recent years. For example, we now know that each cortex has more connections between its cells than there are stars in the Milky Way Galaxy.



▷ **Mapping the brain**
The functional areas are mapped on the outside of the brain. Different parts of the brain cooperate and interact with each other to produce other functions, such as planning or operating machinery.